

Material Selection Recommendation System

A DBMS-Based Decision Support System for Engineering Material Selection

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What is the Material Selection System?

Material Selection Recommendation System

A personalized engineering material selection platform that helps users identify suitable materials based on application-specific requirements.

Core Capabilities

- User registration and secure login
- Application-specific requirement selection
- Dynamic material property database
- Constraint-based material filtering
- Weighted material ranking
- Material comparison
- Saved projects and recommendation history

Why this Project?

- Selecting an appropriate material requires consideration of multiple engineering factors.
- Different applications require different combinations of material properties.
- Manual searching through material databases and handbooks can be time-consuming.
- Engineers need to balance performance, manufacturability, availability and cost.
- A database-driven system can organize this information and automate the initial selection process.

Project Goal

Develop a practical decision-support system that dynamically identifies and ranks suitable engineering materials according to user-defined requirements.

Proposed Features

User Management

- Registration and Login
- User profile
- Project management
- Search history
- Saved recommendations

Material Database

- Multiple material categories
- Dynamic material properties
- Manufacturing compatibility
- Cost and availability
- Data-source tracking

Requirement Management

- Application-specific requirements
- Dynamic property selection
- Mandatory / preferred constraints
- Weighted preferences

Recommendation Engine

- Constraint filtering
- Weighted ranking
- Material comparison
- Recommendation justification

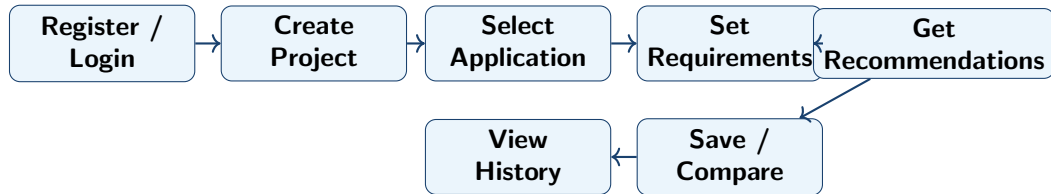
System Architecture



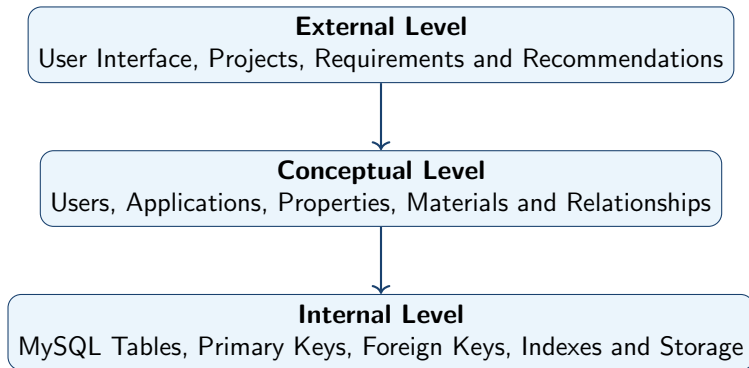
Technology Stack

Frontend: React **Backend:** Node.js + Express.js **Database:** MySQL **ORM:** Sequelize

User Workflow



Three-Schema Architecture



System Modules

Module 1	Module 2	Module 3	Module 4
User Management Registration Login Profile History	Requirement Management Applications Properties Constraints Preferences	Material Database Materials Properties Processes Sources	Recommendation Filtering Ranking Comparison Justification

Module 1 – User & Project Management

User Account

- Registration
- Login and Logout
- Profile management
- Secure password storage

Selection Projects

- Create multiple projects
- Store project requirements
- Save recommendations
- View previous selections

Example

A user can maintain separate projects such as *Heat Sink Design*, *Structural Bracket* and *Bearing Selection*.

User & Project Database Schema

USERS

- User_ID (**PK**)
- Name
- Email (**UNIQUE**)
- Password_Hash
- Created_At
- Last_Login

SELECTION_PROJECT

- Project_ID (**PK**)
- User_ID (**FK**)
- Project_Name
- Application_ID (**FK**)
- Created_At
- Updated_At

USERS 1 : M SELECTION_PROJECT

Module 2 – Dynamic Requirements

User requirements are application-specific.

Mechanical

- Strength
- Stiffness
- Hardness
- Wear resistance
- Fatigue
- Density

Thermal

- Operating temperature
- Service temperature
- Thermal conductivity
- Thermal expansion

Environmental

- Corrosion resistance
- Chemical resistance
- Moisture exposure
- UV exposure
- Marine environment

Other Requirements

Manufacturing process — Budget — Availability — Property priority

Application-Specific Requirements

Different applications require different material properties.

Application	Primary Property	Secondary Property	Other Constraint
Structural Bracket	Yield Strength	Density	Corrosion Resistance
Heat Sink	Thermal Conductivity	Density	Service Temperature
Bearing	Hardness / Wear	Fatigue Strength	Temperature
Electrical Insulator	Dielectric Strength	Electrical Resistivity	Temperature
Chemical Tank	Chemical Resistance	Strength	Cost

Key Idea

The system loads relevant properties according to the selected application instead of forcing every application to use the same fixed requirements.

Dynamic Requirement Database Schema

APPLICATION

- Application_ID (**PK**)
- Application_Name
- Description

PROPERTY

- Property_ID (**PK**)
- Property_Name
- Category
- Data_Type
- Default_Unit

APPLICATION_PROPERTY

- Application_ID (**FK**)
- Property_ID (**FK**)
- Importance_Weight
- Default_Operator
- Default_Target

PROJECT_REQUIREMENT

- Requirement_ID (**PK**)
- Project_ID (**FK**)
- Property_ID (**FK**)
- Operator

Module 3 – Material Database

Material Categories

Metals — Polymers — Ceramics — Composites

Material Information

- Material name
- Grade
- Category
- Description
- Cost
- Availability

Dynamic Properties

- Mechanical properties
- Thermal properties
- Electrical properties
- Environmental properties
- Physical properties

Material Database Schema

MATERIAL

- Material_ID **(PK)**
- Material_Name
- Category_ID **(FK)**
- Grade
- Description
- Source_ID **(FK)**
- Last_Updated

MATERIAL_CATEGORY

- Category_ID **(PK)**
- Category_Name

MATERIAL_PROPERTY

- Material_ID **(FK)**
- Property_ID **(FK)**
- Value
- Unit

DATA_SOURCE

- Source_ID **(PK)**
- Source_Name
- Source_Type
- Reference
- Access_Date

Dynamic Material Properties

The database does not require a fixed column for every possible property.

Material	Property	Value	Unit
Aluminium 6061-T6	Density	2700	kg/m ³
Aluminium 6061-T6	Yield Strength	276	MPa
Aluminium 6061-T6	Thermal Conductivity	167	W/mK
Steel 304	Yield Strength	215	MPa
Steel 304	Thermal Conductivity	16.2	W/mK

Advantage

New material properties can be added to the database without redesigning the entire material table.

Manufacturing & Economic Constraints

Manufacturing Process

- Casting
- Forging
- Machining
- Injection Molding
- Extrusion
- Additive Manufacturing
- Sheet Metal Forming

Economic Factors

- Material cost
- Budget range
- Availability
- Production volume
- Supplier information

Selection Logic

A material should satisfy both the required engineering properties and practical manufacturing/economic constraints.

Manufacturing Database Schema

MANUFACTURING_PROCESS

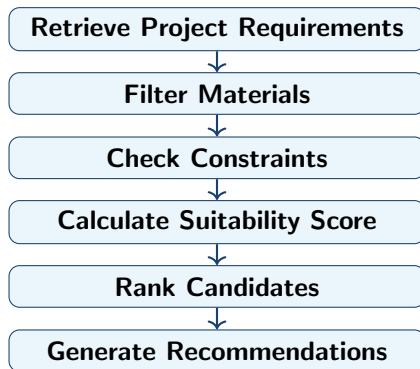
- Process_ID (**PK**)
- Process_Name
- Description

MATERIAL_PROCESS

- Material_ID (**FK**)
- Process_ID (**FK**)
- Suitability

MATERIAL_COST

- Cost_ID (**PK**)
- Material_ID (**FK**)
- Cost_Per_Kg
- Currency
- Source
- Last_Updated



Material Selection Algorithm

Step 1 – Retrieve Requirements

Obtain the selected application's relevant properties and the user's target values.

Step 2 – Constraint Filtering

- Remove materials that violate mandatory requirements.
- Check mechanical, thermal, environmental and manufacturing constraints.

Step 3 – Weighted Ranking

Remaining materials are scored according to the user's property priorities.

Conceptual Suitability Score

$$Score = \sum_{i=1}^n w_i \times M_i$$

where w_i represents the importance weight and M_i represents the normalized property match.

Example Selection Requirements

Application: Heat Sink

Thermal Conductivity	$\geq 150 \text{ W/mK}$
Density	$\leq 3000 \text{ kg/m}^3$
Service Temperature	$\geq 100^\circ\text{C}$
Cost	$\leq 1000/\text{kg}$

User Priorities

Thermal Conductivity	5/5
Density	4/5
Temperature	4/5
Cost	3/5

Result

The system filters unsuitable materials and ranks the remaining candidates according to the user's priorities.

Recommendation Output

- Material name
- Suitability score
- Rank
- Matched properties
- Failed / weak requirements
- Manufacturing compatibility
- Estimated cost
- Data source

Material	Strength	Density	Thermal	Cost	Score
Aluminium 6061-T6	Medium	Low	High	Medium	91%
Copper	Medium	High	Very High	High	87%
Aluminium 6063	Medium	Low	High	Medium	82%

Illustrative values for demonstration

Recommendation Database Schema

RECOMMENDATION

- Recommendation_ID (**PK**)
- Project_ID (**FK**)
- Material_ID (**FK**)
- Suitability_Score
- Rank
- Reason
- Generated_At

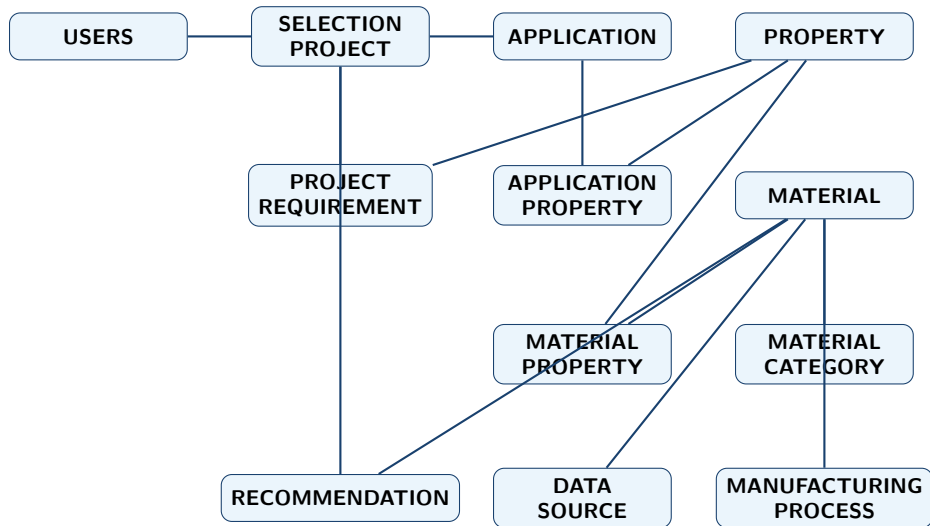
SEARCH_HISTORY

- History_ID (**PK**)
- User_ID (**FK**)
- Project_ID (**FK**)
- Search_Date

Purpose

Store previous selections and recommendations so users can revisit them later.

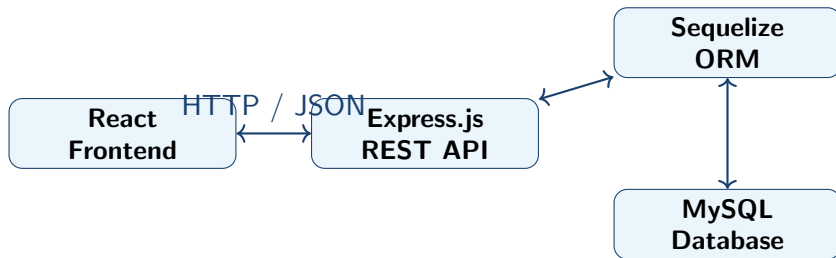
Overall ER Model



Database Relationships

Relationship	Cardinality	Purpose
User → Project	1 : M	User can create multiple projects
Project → Requirement	1 : M	Project can contain multiple requirements
Application → Property	M : M	Applications use different properties
Material → Property	M : M	Materials have multiple properties
Material → Category	M : 1	Each material belongs to a category
Project → Recommendation	1 : M	Project generates recommendations
Material → Process	M : M	Materials support multiple processes
Material → Source	M : 1	Material data has a source

REST API Architecture



Example API Operations

GET Materials — POST Requirements — GET Recommendations — GET Project History

Data Sources & Data Reliability

Material information can be collected and maintained from reliable sources such as:

- Manufacturer datasheets
- Engineering material databases
- Standards and technical references
- Public materials-data repositories

Source Tracking

Each material record can store its source, reference and last-updated date.

Important

The prototype database will use a curated set of verified material records. External databases should not be automatically scraped without checking their licensing and usage conditions.

Future Scope

- Expand the engineering material database
- Add more application templates
- Machine-learning-based recommendation
- Advanced multi-objective optimization
- Sustainability and environmental impact analysis
- Supplier and real-time availability information
- Advanced graphical property comparison
- CAD / CAE integration
- Mobile application
- Cloud-based database

Conclusion

The Material Selection Recommendation System provides a structured DBMS-based approach to engineering material selection.

- Provides personalized user accounts and project history.
- Supports application-specific and dynamic material requirements.
- Maintains a structured database of materials and properties.
- Considers engineering, manufacturing and economic constraints.
- Filters and ranks materials using weighted requirements.
- Provides recommendations, comparisons and selection justification.

Thank You